PROOF OF CONCEPT

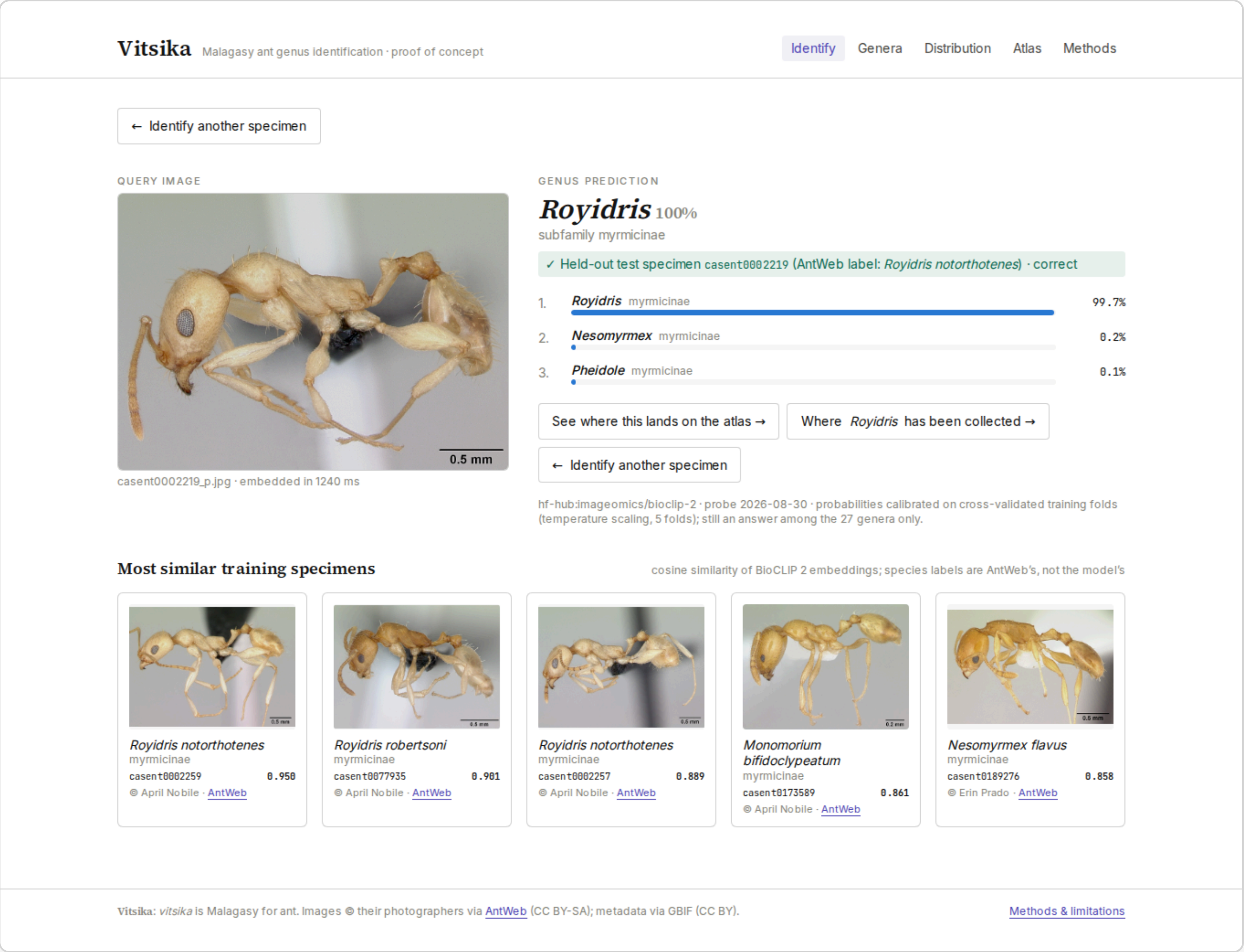
Vitsika

Malagasy ant identification from a single photograph

vitsika is Malagasy for ant. Genus-level, 27 genera, built on public data and a frozen public model.

vitsika.lucas-ralambo.com

Lucas R. · Data Scientist · Antananarivo
2026-08-30



Genus-level identification is scarce expertise - a local first pass multiplies it

- AntWeb's GBIF dataset held **141,430** Malagasy ant records on 2026-08-26, **6,995** with images, across **60** imaged genera (GBIF occurrence API; doi:10.15468/wqmjjt). Naming even the genus takes a specialist and a microscope.
- Three decades of collecting, imaging and determination by the California Academy of Sciences and Malagasy field teams make this data trainable at all. MBC's Phase III roadmap: ‘with training, more steps can be done in Madagascar’ (madagascarbio.org/monitoring.html).
- Vitsika demonstrates the division of labour: a photo-to-genus first pass runs in under a second on an ordinary computer in Antananarivo - fast enough to process a full night's camera-trap haul locally, with standard optimizations still to apply. It clears the routine identifications and flags what it cannot place, so specialist time goes to the ambiguous and the possibly new.

One AntWeb profile shot per genus in scope (27).

From GBIF to 1,236 clean images in 27 genera

141,430
AntWeb records for Madagascar on GBIF (doi:10.15468/wqmjjt); 6,995 with an image

→

4,354
specimens, 39 genera: workers, profile view, one image each, ≥ 10 per genus

→

1,297
images obtained (29.8 %): 1,160 Wikimedia Commons + 137 GBIF cache

→

1,236
images, **27 genera**, 8 subfamilies; 986 train / 250 held-out test

Finding 1: GBIF's image cache is empty for AntWeb

4,215 of 4,354 cache requests returned 404 (96.8 %), uniformly across years and record age. antweb.org blocks scripts (Cloudflare) and datacenter IPs (403), including GBIF's own fetcher. The working source is the 2008-2014 bulk upload to Wikimedia Commons (33,081 files), matched by specimen code.

Finding 2: caste is not in GBIF's interpreted fields

GBIF maps `sex` to Male / Female / Other, so every worker arrives as null: 4,540 of 5,857 specimens. The caste has to be read from the verbatim record (`dwc:sex`): 4,539 workers, 673 queens, 640 males, and only workers are kept.

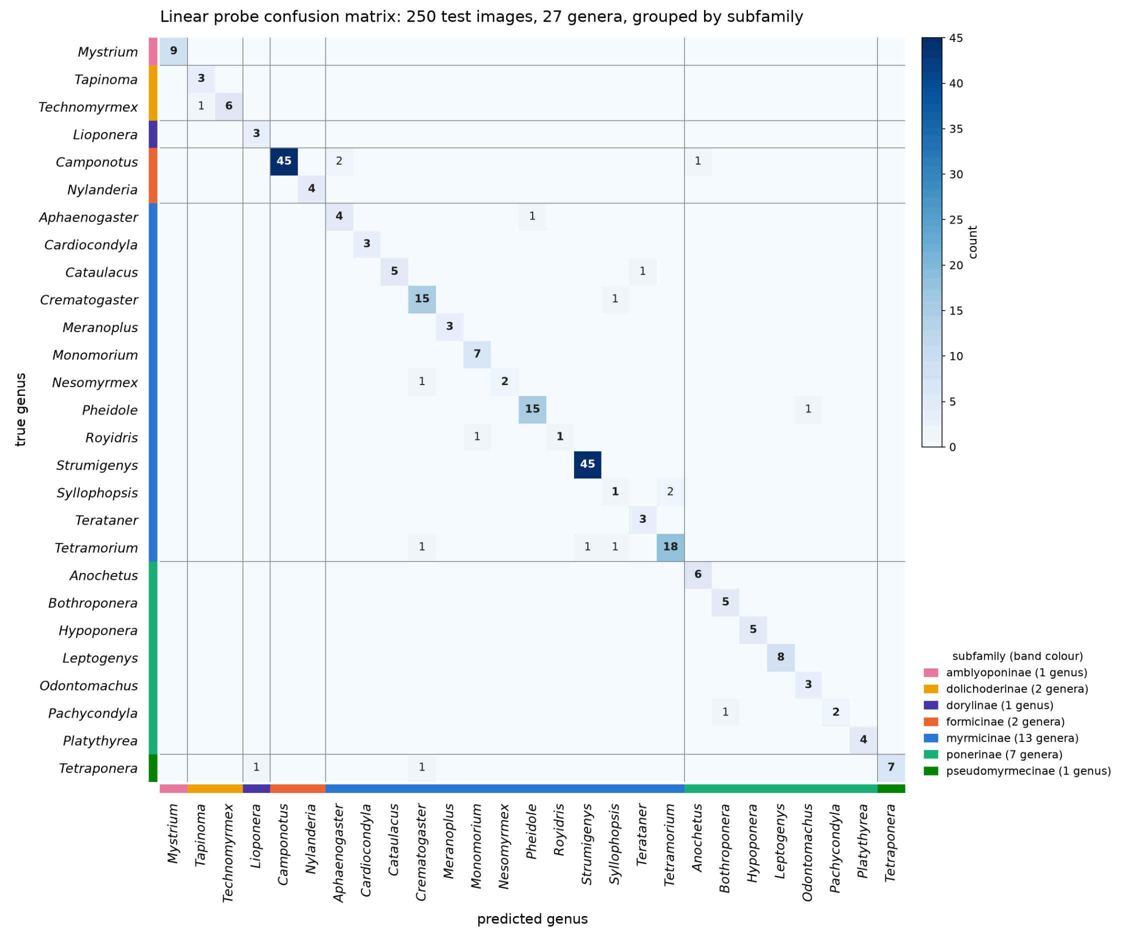
Linear probe on frozen BioCLIP 2 embeddings: 92.8 % top-1 on 250 held-out specimens

METHOD	TOP-1	TOP-3	MACRO-F1
Zero-shot, “a photo of {genus}, a genus of ant”	74.4 %	94.0 %	0.590
Linear probe (deployed)	**92.8 %**	**98.4 %**	**0.885**
Nearest neighbour, cosine	91.6 %	93.2 %	0.834

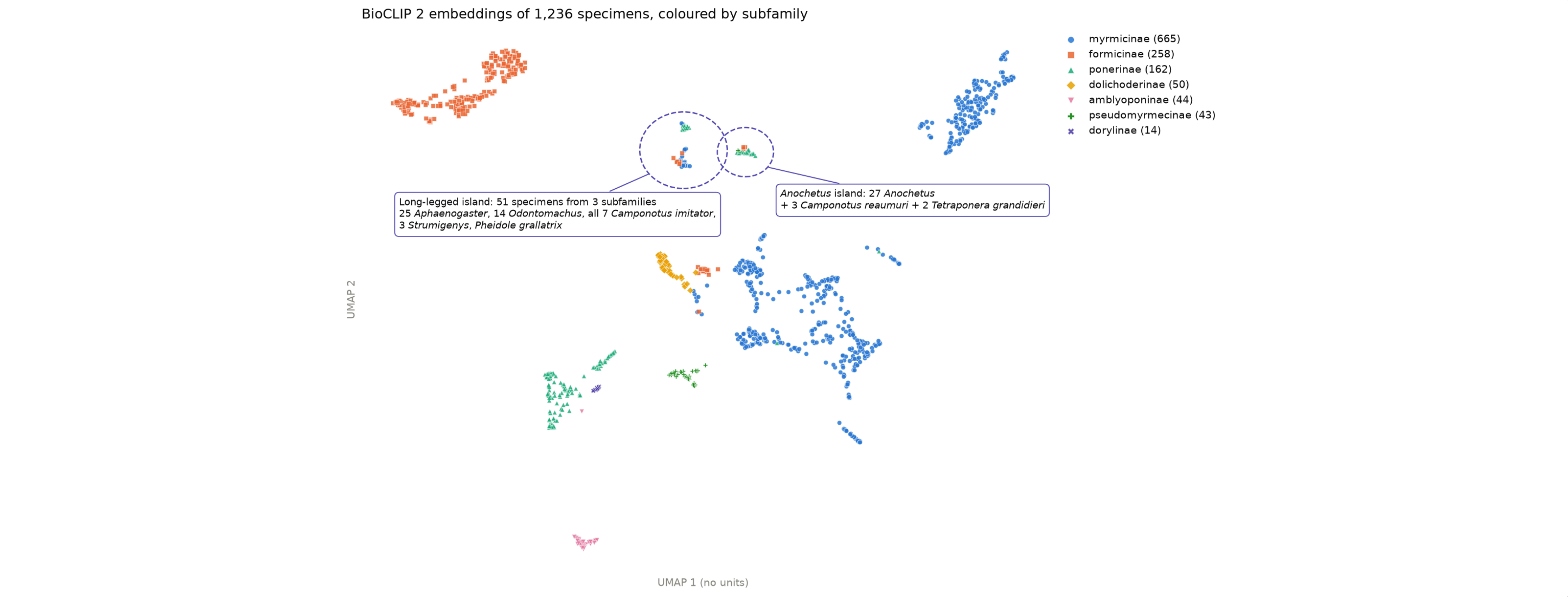
986 train / 250 test, 27 genera, split grouped by specimen, stratified by genus (seed 42). Bare genus name as the zero-shot prompt: 32.8 % top-1. Embeddings: `imageomics/bioclip-2` ViT-L/14, 768-d, no fine-tuning.

Full evaluation, error analysis and reproducible pipeline: `github.com/LucasRal/vitsika`

> 18 errors: 12 genuine look-alikes or species unseen in training (*Syllophopsis* vs *Tetramorium*, *Royidris* vs *Monomorium*), 3 poor photos, 2 odd angles, 1 unexplained. Almost every confusion stays inside a subfamily block.



The embedding space recapitulates taxonomy, with body-plan islands



Subfamilies form clean clusters; where the model "errs", it groups *Camponotus imitator* with long-legged *Aphaenogaster* and *Odontomachus*, a resemblance a taxonomist would recognise, not noise.

NEXT STEPS

Three things this codebase already supports

1

Complete the image set

70.2 % of the manifest's images are missing. `05_download_antweb.py` + `dataset_full.csv` are ready for a residential IP or an AntWeb bulk export; re-thresholding brings *Vitsika*, *Tanipone*, *Carebara* and 9 more genera into scope (39 in total).

2

Species-grouped split and an open-set reject

Group the split by species (one flag in `03_build_dataset.py`) and turn the nearest-neighbour similarity the API already returns into a "none of the 27" reject (probability calibration: done, live since 2026-08-30).

3

Multi-view, multi-caste embeddings

The harvest already holds 6,634 head and 5,127 dorsal media rows plus 673 queens and 640 males; `config.yaml` (`views_priority`, `castes`) and `06_embed.py` make view-wise embedding and late fusion a configuration change.

Lucas R. · Data Scientist, Antananarivo · aina@lucas-ralambo.com · vitsika.lucas-ralambo.com · github.com/LucasRal/vitsika